

SECTION 7.A

Determine Exact Trigonometric Values

(Unit Circle)

Learning Objectives

- I can determine exact values of sine, cosine, and tangent for special angles.
- I can determine the sign of trig functions using quadrant patterns.
- I can evaluate trig functions for negative angles.
- I can evaluate trig functions for angles greater than 2π using coterminal angles.

Introduction

The unit circle allows us to determine exact values of sine, cosine, and tangent without a calculator. Instead of memorizing disconnected numbers, we use structure and patterns.

This skill is foundational.

Just like factoring reveals x-intercepts in algebra, knowing exact trig values allows us to:

- Solve equations like $\sin \theta = \frac{1}{2}$
- Identify where trig graphs cross axes
- Graph sine and cosine functions accurately
- Model periodic motion and circular systems

Everything we do later with trigonometry depends on understanding these exact values.

Key Vocabulary

Unit Circle: A circle centered at the origin with radius 1.

Cosine: The x-coordinate of a point on the unit circle.

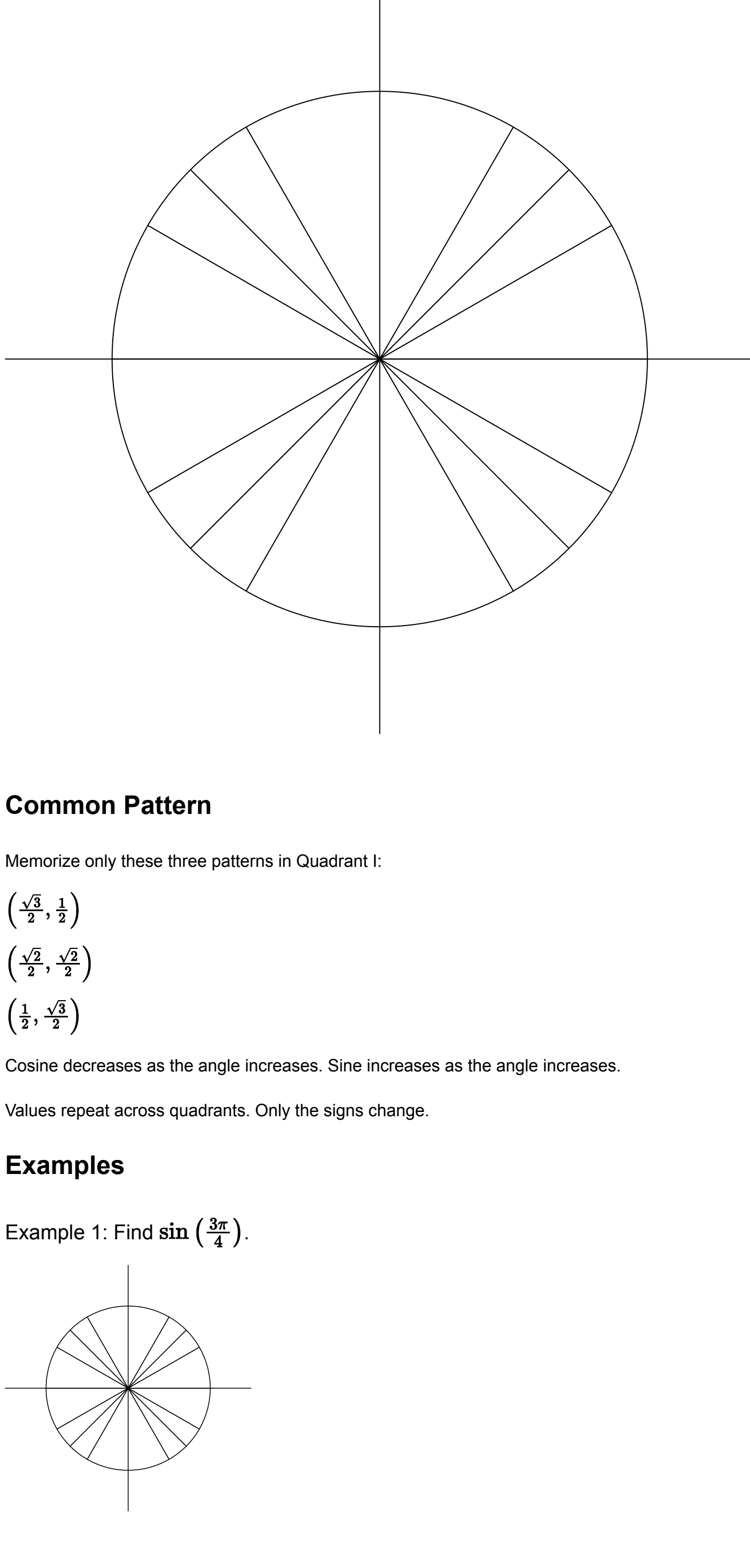
Sine: The y-coordinate of a point on the unit circle.

Tangent: $\tan \theta = \frac{\sin \theta}{\cos \theta}$

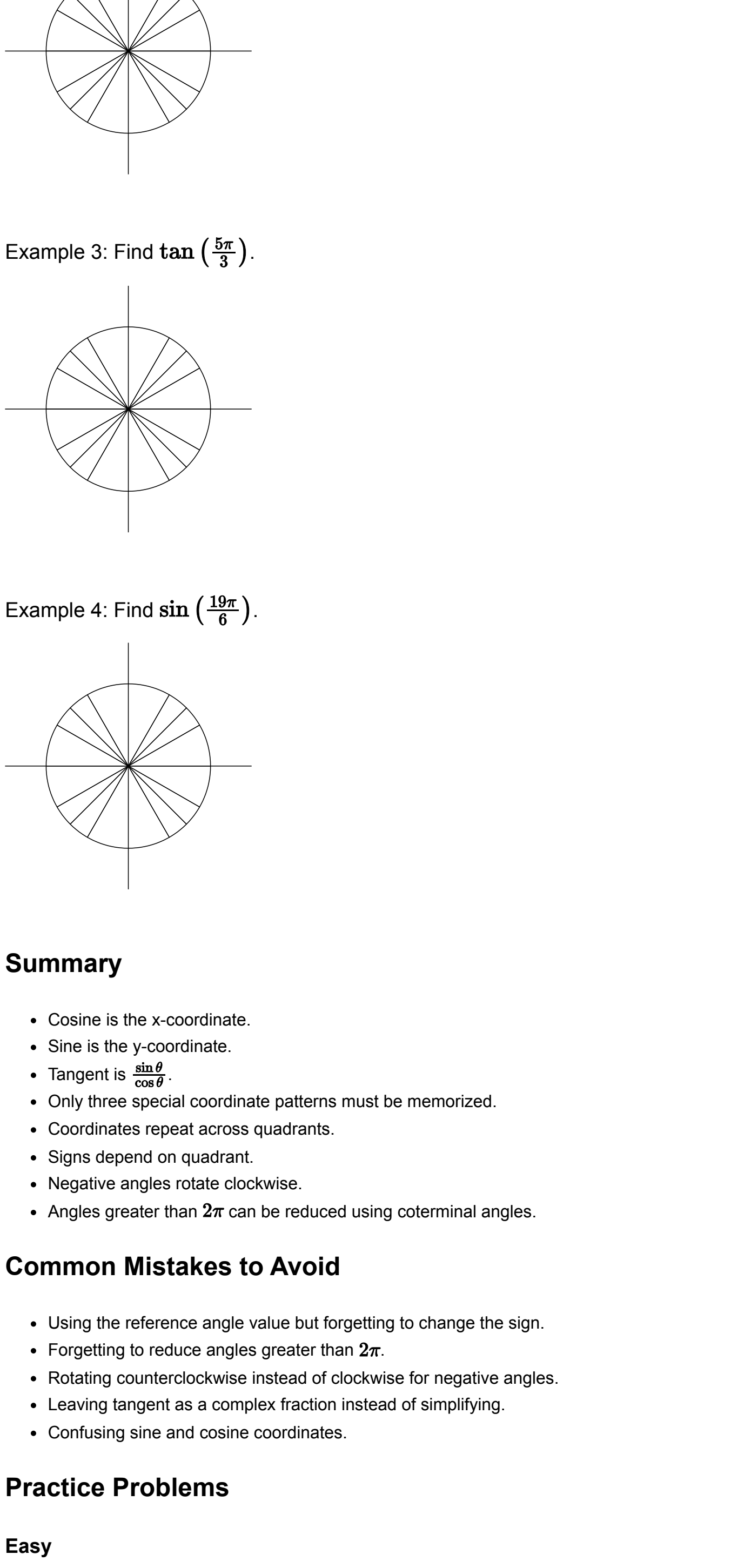
Reference Angle: The acute angle formed between the terminal side and the x-axis.

Coterminal Angles: Angles that differ by multiples of 2π .

Blank Unit Circle (Practice 1)



Blank Unit Circle (Practice 2)



Common Pattern

Memorize only these three patterns in Quadrant I:

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

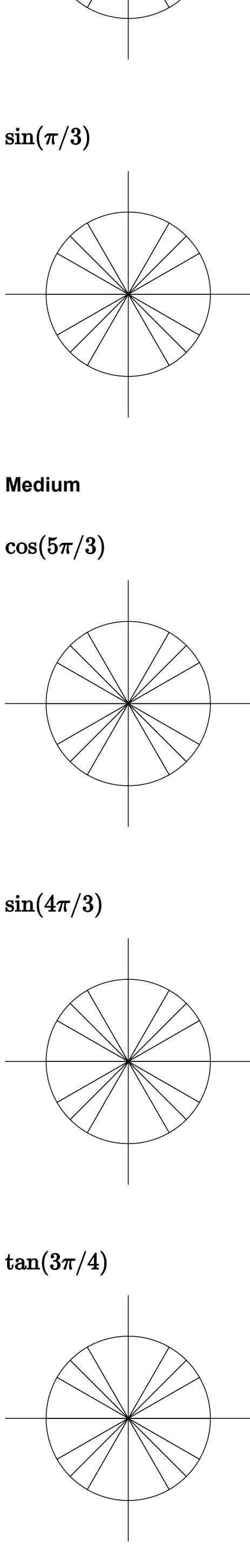
$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

Cosine decreases as the angle increases. Sine increases as the angle increases.

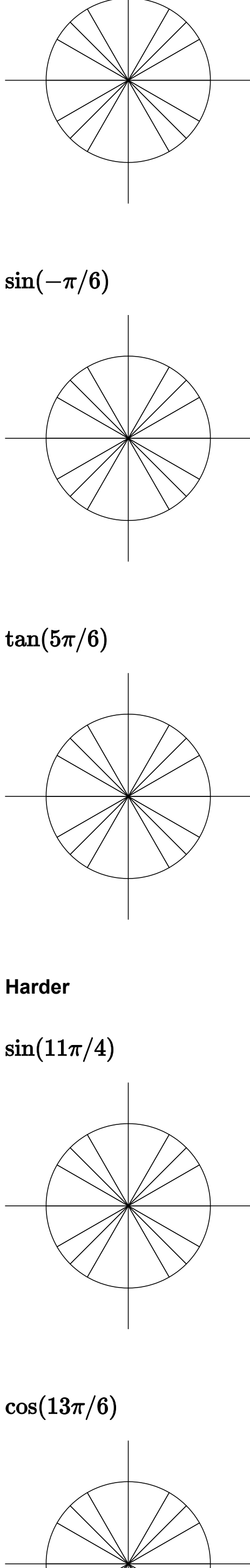
Values repeat across quadrants. Only the signs change.

Examples

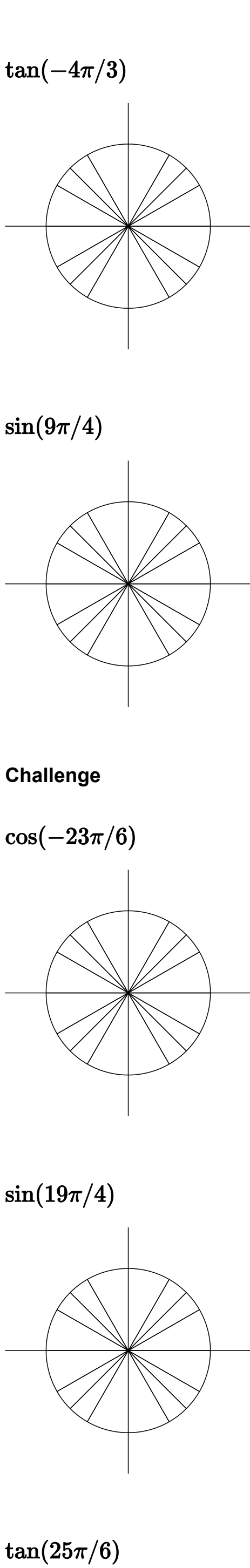
Example 1: Find $\sin\left(\frac{3\pi}{4}\right)$.



Example 2: Find $\cos\left(-\frac{\pi}{6}\right)$.



Example 3: Find $\tan\left(\frac{5\pi}{3}\right)$.



Example 4: Find $\sin\left(\frac{19\pi}{6}\right)$.



Summary

- Cosine is the x-coordinate.
- Sine is the y-coordinate.
- Tangent is $\frac{\sin \theta}{\cos \theta}$.
- Only three special coordinate patterns must be memorized.
- Coordinates repeat across quadrants.
- Signs depend on quadrant.
- Negative angles rotate clockwise.
- Angles greater than 2π can be reduced using coterminal angles.

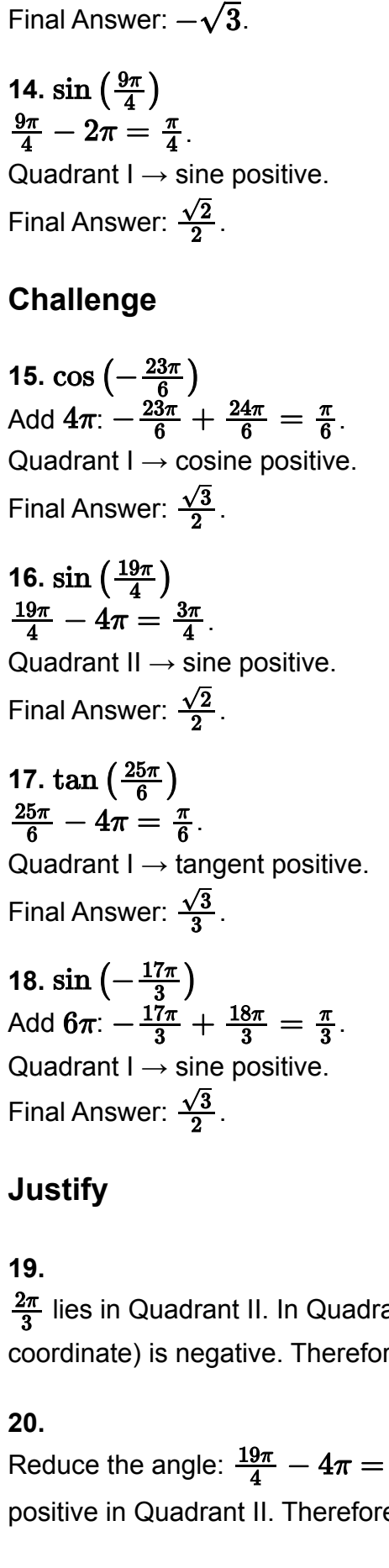
Common Mistakes to Avoid

- Using the reference angle value but forgetting to change the sign.
- Forgetting to reduce angles greater than 2π .
- Rotating counterclockwise instead of clockwise for negative angles.
- Leaving tangent as a complex fraction instead of simplifying.
- Confusing sine and cosine coordinates.

Practice Problems

Easy

$\sin(\pi/4)$



$\cos(\pi/3)$

$\tan(\pi/6)$

$\sin(\pi/3)$

Medium

$\cos(5\pi/3)$

$\sin(4\pi/3)$

$\tan(3\pi/4)$

$\cos(7\pi/6)$

$\sin(-\pi/6)$

$\tan(5\pi/6)$

Harder

$\sin(11\pi/4)$

$\cos(13\pi/6)$

$\tan(-4\pi/3)$

$\sin(9\pi/4)$

Challenge

$\cos(-23\pi/6)$

$\sin(19\pi/4)$

$\tan(25\pi/6)$

$\sin(-17\pi/3)$

Justify

19. Justify why $\sin(2\pi/3)$ is positive but $\cos(2\pi/3)$ is negative.
20. Justify how you would determine $\sin(19\pi/4)$ without using a diagram. Include coterminal angle reasoning, reference angle, and quadrant.

Solutions

Easy

1. $\sin\left(\frac{\pi}{4}\right)$
Step 1: $\frac{\pi}{4}$ is in Quadrant I.
Step 2: Reference angle = $\frac{\pi}{4}$.
Step 3: Sine is positive in Quadrant I.
Final Answer: $\frac{\sqrt{2}}{2}$.
2. $\cos\left(\frac{\pi}{3}\right)$
Quadrant I $\rightarrow$ cosine is positive.
Reference value for $\frac{\pi}{3}$ is $\frac{1}{2}$.
Final Answer: $\frac{1}{2}$.
3. $\tan\left(\frac{\pi}{6}\right)$
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$.
 $\frac{1}{2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$.
Final Answer: $\frac{\sqrt{3}}{3}$.
4. $\sin\left(\frac{\pi}{3}\right)$
Quadrant I $\rightarrow$ sine positive.
Final Answer: $\frac{\sqrt{3}}{2}$.

Medium

5. $\cos\left(\frac{5\pi}{3}\right)$
Quadrant IV $\rightarrow$ cosine positive.
Reference angle = $\frac{\pi}{3}$.
Final Answer: $\frac{1}{2}$.
6. $\sin\left(\frac{4\pi}{3}\right)$
Quadrant III $\rightarrow$ sine negative.
Reference angle = $\frac{\pi}{3}$.
Final Answer: $-\frac{\sqrt{3}}{2}$.
7. $\tan\left(\frac{3\pi}{4}\right)$
Quadrant II $\rightarrow$ tangent negative.
Reference angle = $\frac{\pi}{4}$.
Final Answer: -1 .
8. $\cos\left(\frac{7\pi}{6}\right)$
Quadrant III $\rightarrow$ cosine negative.
Reference angle = $\frac{\pi}{6}$.
Final Answer: $-\frac{\sqrt{3}}{2}$.
9. $\sin\left(-\frac{\pi}{6}\right)$
Negative angle $\rightarrow$ rotate clockwise to Quadrant IV.
Sine negative in Quadrant IV.
Final Answer: $-\frac{1}{2}$.
10. $\tan\left(\frac{5\pi}{6}\right)$
Quadrant II $\rightarrow$ tangent negative.
Reference angle = $\frac{\pi}{6}$.
Final Answer: $-\frac{\sqrt{3}}{3}$.

Harder

11. $\sin\left(\frac{11\pi}{4}\right)$
 $\frac{11\pi}{4} - 2\pi = \frac{3\pi}{4}$.
Quadrant II $\rightarrow$ sine positive.
Final Answer: $\frac{\sqrt{2}}{2}$.
12. $\cos\left(\frac{13\pi}{6}\right)$
 $\frac{13\pi}{6} - 2\pi = \frac{\pi}{6}$.
Quadrant I $\rightarrow$ cosine positive.
Final Answer: $\frac{\sqrt{3}}{2}$.
13. $\tan\left(-\frac{4\pi}{3}\right)$
Add 2π : $-\frac{4\pi}{3} + 2\pi = \frac{2\pi}{3}$.
Quadrant II $\rightarrow$ tangent negative.
Reference angle = $\frac{\pi}{3}$.
Final Answer: $-\sqrt{3}$.
14. $\sin\left(\frac{9\pi}{4}\right)$
 $\frac{9\pi}{4} - 2\pi = \frac{\pi}{4}$.
Quadrant I $\rightarrow$ sine positive.
Final Answer: $\frac{\sqrt{2}}{2}$.

Challenge

15. $\cos\left(-\frac{23\pi}{6}\right)$
Add 4π : $-\frac{23\pi}{6} + \frac{24\pi}{6} = \frac{\pi}{6}$.
Quadrant I $\rightarrow$ cosine positive.
Final Answer: $\frac{\sqrt{3}}{2}$.
16. $\sin\left(\frac{19\pi}{4}\right)$
 $\frac{19\pi}{4} - 4\pi = \frac{3\pi}{4}$.
Quadrant II $\rightarrow$ sine positive.
Final Answer: $\frac{\sqrt{2}}{2}$.
17. $\tan\left(\frac{25\pi}{6}\right)$
 $\frac{25\pi}{6} - 4\pi = \frac{\pi}{6}$.
Quadrant I $\rightarrow$ tangent positive.
Final Answer: $\frac{\sqrt{3}}{3}$.
18. $\sin\left(-\frac{17\pi}{3}\right)$
Add 6π : $-\frac{17\pi}{3} + \frac{18\pi}{3} = \frac{\pi}{3}$.
Quadrant I $\rightarrow$ sine positive.
Final Answer: $\frac{\sqrt{3}}{2}$.

Justify

19. $\frac{2\pi}{3}$ lies in Quadrant II. In Quadrant II, sine (the y-coordinate) is positive and cosine (the x-coordinate) is negative. Therefore $\sin\left(\frac{2\pi}{3}\right) > 0$ and $\cos\left(\frac{2\pi}{3}\right) < 0$.
20. Reduce the angle: $\frac{19\pi}{4} - 4\pi = \frac{3\pi}{4}$. $\frac{3\pi}{4}$ lies in Quadrant II with reference angle $\frac{\pi}{4}$. Sine is positive in Quadrant II. Therefore $\sin\left(\frac{19\pi}{4}\right) = \frac{\sqrt{2}}{2}$.